

- (b) Table 5.1 compares features of European bison and American bison.

Table 5.1

feature	European bison, <i>B. bonasus</i>	American bison, <i>B. bison</i>
type of food	grass and higher vegetation such as leaves of bushes and trees	grass and low vegetation
height/m	2.1	2.0
maximum mass/kg	1000	1270

Assess the relative importance of natural selection and genetic drift in producing the different heights and masses of the two species of bison.

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- (c) The European bison has a nuclear genome that is very similar to that of the American bison. The European bison has a mitochondrial genome that is more similar to that of wild cattle of the genus *Bos* than to the American bison.

Discuss what this implies about the evolutionary history of the European bison.

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- (d) Outline how practical techniques could be used to test the hypothesis that farmed cattle are closely related to European bison.

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